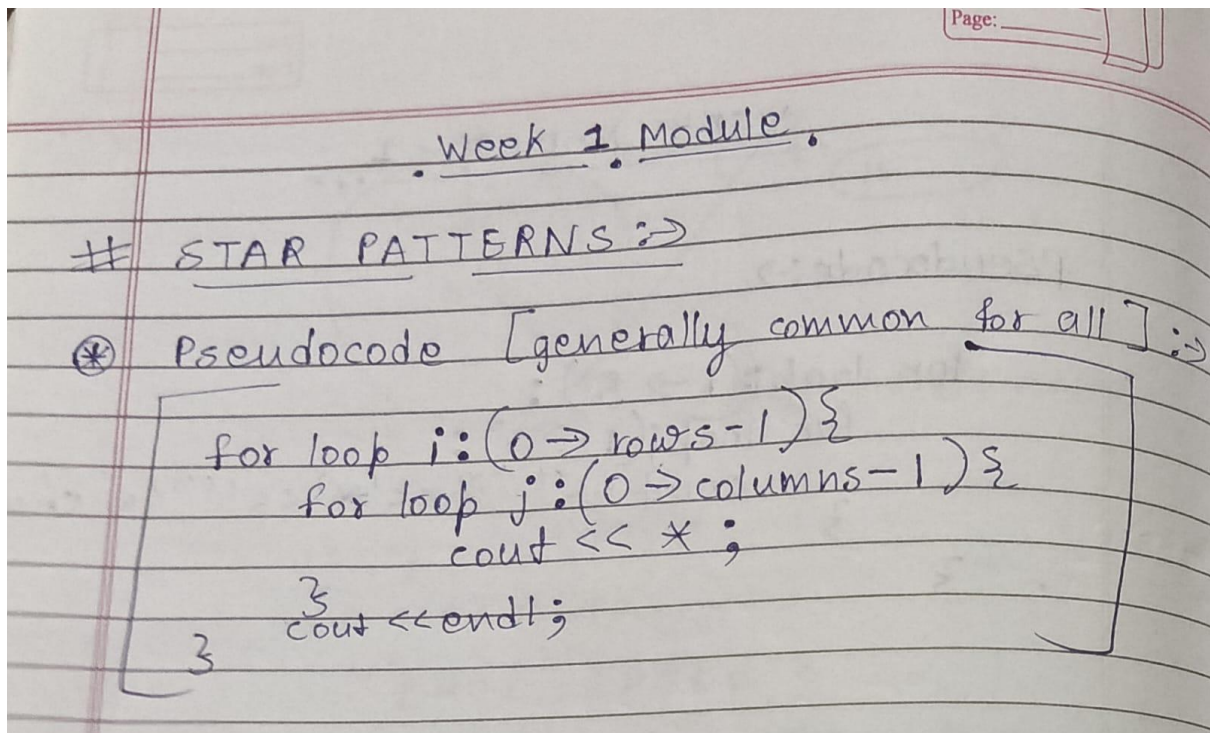


Week 1 Module

PseudoCode:-

Star Patterns



1. Square

① Square pattern :-

```
for (int i=0; i<5; i++) {  
    for (int j=0; j<5; j++) {  
        cout << " " << "*" << " ";  
    }  
    cout << endl;  
}
```

* {i loop} $\Rightarrow$ To control the rows in pattern
 $i: (0 \rightarrow 5)$ { 5 rows in sq. pattern }
{j loop} $\Rightarrow$ To control the columns in pattern
 $j: (0 \rightarrow 5)$ { 5 column in sq. pattern }
cout << endl; { in outer loop to enter the new row }

OUTPUT:-

```
Week 1 Module.cpp > main()  
1  #include<iostream>  
2  using namespace std;  
3  
4  int main(){  
5      for (int i=0; i<5; i++){ //row controller  
6  
7          for (int j=0; j<5; j++){ //column controller  
8  
9              cout<<" "<<"*"<<" "; //print the star  
10         }  
11  
12         cout<<endl; //new line after each row  
13     }  
14     return 0;  
15 }
```

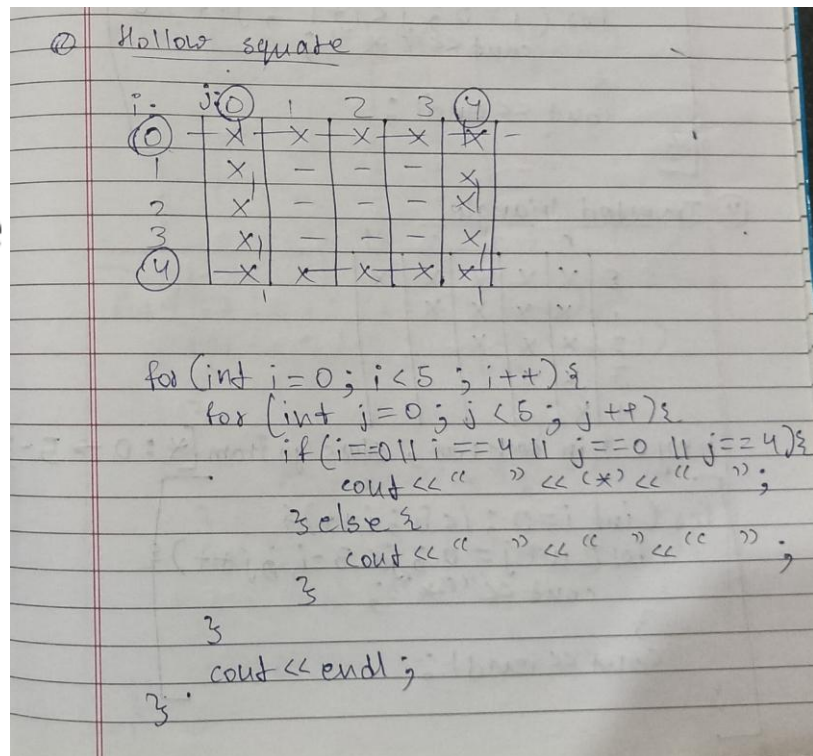
```
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp  
● $ g++ Week\ 1\ Module.cpp -o main.exe && ./main.exe  
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *  
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp  
○ $
```

2. Hollow Square

* *

* *

* *



OUTPUT:-

```

Week 1 Module.cpp X
Week 1 Module.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      for (int i=0; i<5; i++){ //row controller
6
7          for (int j=0; j<5; j++){ //column controller
8
9              if (i==0 || i==4 || j==0 || j==4){
10
11                  cout<<" "<<"*"<<" "; //print the star
12              }else{
13                  cout<<" "<<" "<<" "; //print the space
14              }
15          }
16
17          cout<<endl; //new line after each row
18      }
19      return 0;
20 }

```

```

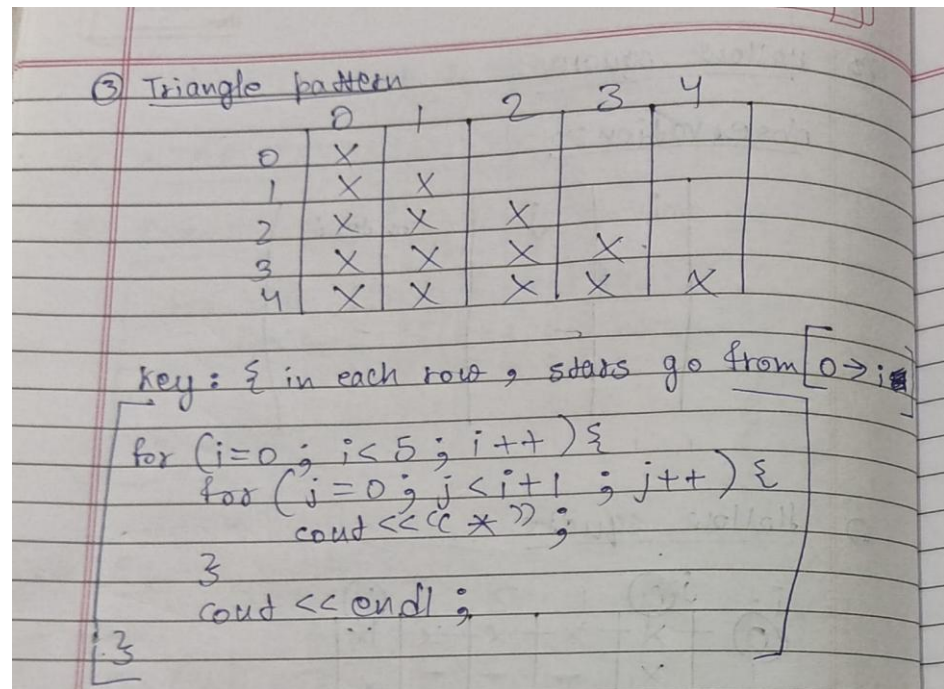
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$ g++ Week\ 1\ Module.cpp -o main.exe && ./main.exe
* * * * *
*       *
*       *
*       *
* * * * *

sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$

```

3. Triangle

```
*
**
***
****
*****
```



OUTPUT:-

```
Week 1 Module.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      for (int i=0; i<5; i++){ //row controller
6          for (int j=0; j<=i; j++){ //column controller
7              cout<<"*"; //print star
8          }
9          cout<<endl; //new line after each row
10     }
11     return 0;
12 }
13
14
15
```

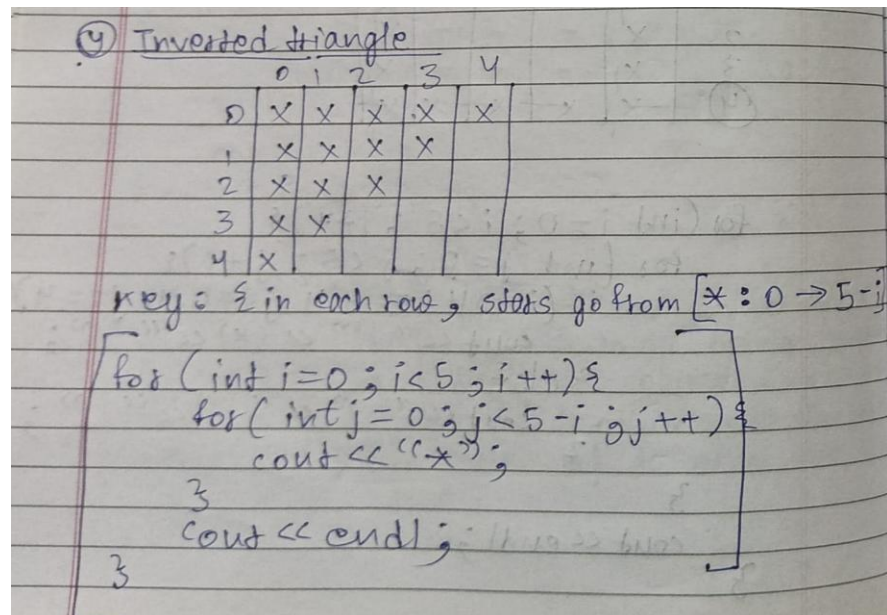
```
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$ g++ Week\ 1\ Module.cpp -o main.exe && ./main.exe
*
* *
* * *
* * * *
* * * * *

sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$
```


4. Inverted Triangle

**

*



OUTPUT:-

```
Week 1 Module.cpp X
Week 1 Module.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      for (int i=0; i<5; i++){ //row controller
6
7          for (int j=0; j<5-i; j++){ //column controller
8
9              cout<<" "<<"x"<<" "; //print star
10         }
11
12         cout<<endl; //new line after each row
13     }
14     return 0;
15 }
```

```
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$ g++ Week\ 1\ Module.cpp -o main.exe && ./main.exe
*****
****
***
**
*

sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$
```

5. Pyramid

```

*
***
*****
*****
*****

```

⑤ pyramid

	0	1	2	3	4	5	6	7	8
0	-	-	-	-	x	-	-	-	-
1	-	-	-	x	x	x	-	-	-
2	-	-	x	x	x	x	x	-	-
3	-	x	x	x	x	x	x	x	-
4	x	x	x	x	x	x	x	x	x

mirrored triangle normal triangle

obs table

Row	spaces	left	right
1	4	1	0
2	3	2	1
3	2	3	2
4	1	4	3
5	0	5	4
i	n-i	i	i-1

```

for (int i=1; i<=n; i++){
    for (x=1; x<=n-i; x++){
        cout<<" ";
    }
    for (y=1; y<=i; y++){
        cout<<"*";
    }
    for (z=1; z<=i-1; z++){
        cout<<"*";
    }
    cout<<"\n";
}

```

5...OUTPUT:-

```
Week 1 Module.cpp X
Week 1 Module.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int n=5; //total rows
6
7      for (int i=0 ; i<n; i++){
8
9          for (int x=0; x<n-i; x++){ //print leading spaces
10             cout<<" "<<" "<<" ";
11         }
12
13         for (int y=0; y<i+1; y++){ //print star (left half)
14             cout<<" "<<"*"<<" ";
15         }
16
17         for (int z=0; z<=i-1; z++){ //print star (right half)
18             cout<<" "<<"*"<<" ";
19         }
20         cout<<endl;
21     }
22
23     return 0;
24 }
```

```
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$ g++ Week\ 1\ Module.cpp -o main.exe && ./main.exe

      *
    * * *
  * * * * *
* * * * * * *
* * * * * * * * *

sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$
```

6. Diamond

```

      *
     ***
    *****
   *********
  ***********
 *****
  *****
   ***
    *
  
```

⑥ Diamond

	1	2	3	4	5	6	7	8	9
1	-	-	-	-	x	-	-	-	-
2	-	-	-	x	x	x	-	-	-
3	-	-	x	x	x	x	x	-	-
4	-	x	x	x	x	x	x	x	-
5	x	x	x	x	x	x	x	x	x
6	x	-	x	x	x	x	x	x	-
7	x	-	-	x	x	x	x	-	-
8	x	-	-	-	x	x	x	-	-
9	x	-	-	-	-	x	-	-	-

6...OUTPUT:-

```

Week 1 Module.cpp ×
Week 1 Module.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n = 5;
6      cout << endl;
7
8      for (int i = 0; i < n; i++) {
9          for (int x = 0; x < n - i - 1; x++) {
10             cout << " ";
11         }
12
13         for (int y = 0; y <= i; y++) {
14             cout << " * ";
15         }
16
17         for (int z = 0; z < i; z++) {
18             cout << " * ";
19         }
20         cout << endl;
21     }
22
23     for (int i = n - 2; i >= 0; i--) {
24         for (int x = 0; x < n - i - 1; x++) {
25             cout << " ";
26         }
27
28         for (int y = 0; y <= i; y++) {
29             cout << " * ";
30         }
31
32         for (int z = 0; z < i; z++) {
33             cout << " * ";
34         }
35         cout << endl;
36     }
37
38     return 0;
39 }
  
```

```

sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$ g++ Week\ 1\ Module.cpp -o main.exe && ./main.exe

      *
     * * *
    * * * * *
   * * * * * * *
  * * * * * * * *
 * * * * * * * * *
* * * * * * * * * *
 * * * * *
  * * * *
   * * *
    * *
     *
      *

sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$
  
```

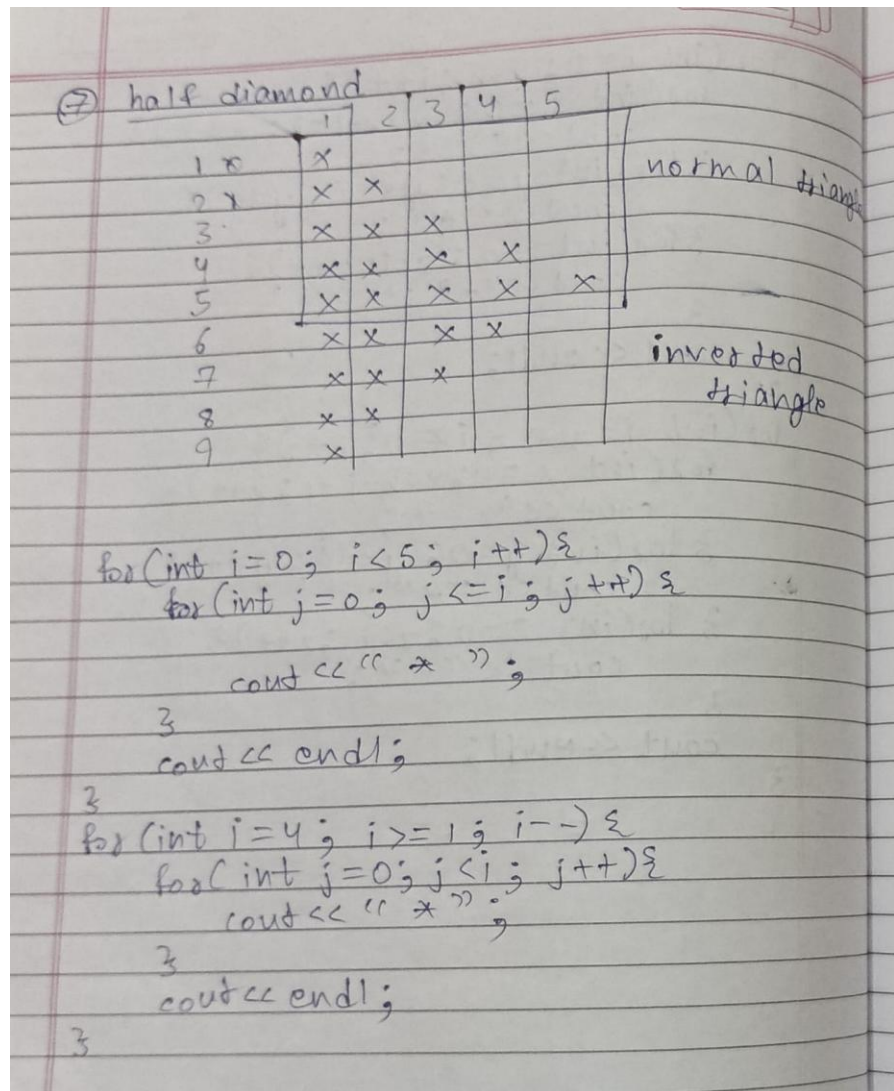

7. Half Diamond

*

**

**

*



7...OUTPUT:-

```

Week 1 Module.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      for (int i=0; i<5; i++){ //for normal triangle
6          for (int j=0; j<=i; j++){
7              cout<<" * ";
8          }
9          cout<<endl;
10     }
11
12     for (int i=4; i>=1; i--){ //for inverted triangle
13         for (int j=0; j<i; j++){
14             cout<<" * ";
15         }
16         cout<<endl;
17     }
18
19     return 0;
20 }
    
```

bash

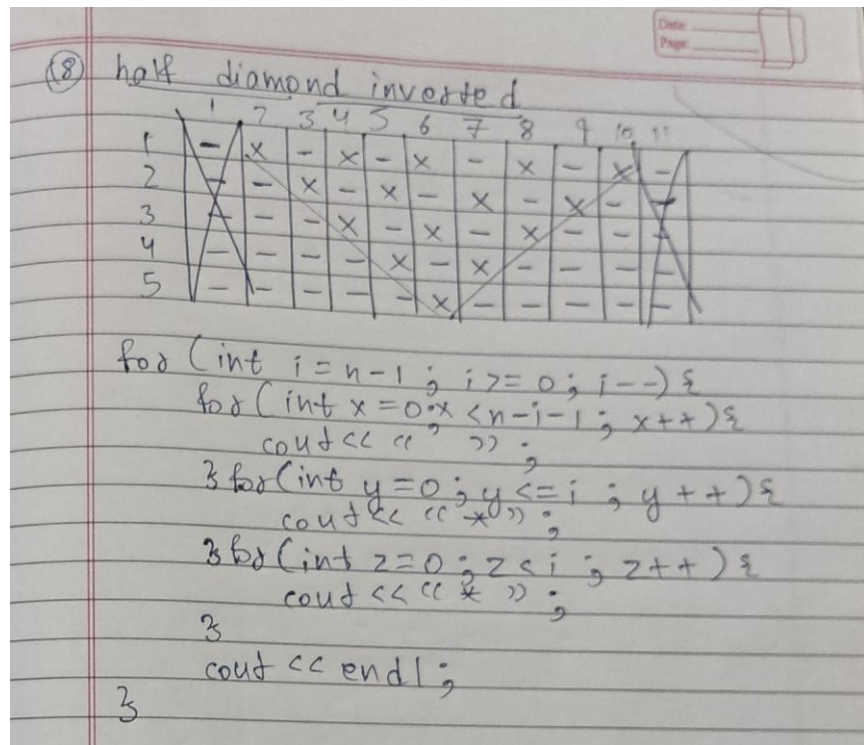
```

sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop
$ g++ Week\ 1\ Module.cpp -o main.exe
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop
$
    
```

8. Half Diamond Inverted

**

*



8...OUTPUT:-

```
Week 1 Module.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n = 5;
6      cout << endl;
7
8      for (int i = n - 1; i >= 0; i--) {
9
10         for (int x = 0; x < n - i - 1; x++) { //spaces
11             cout << " ";
12         }
13
14         for (int y = 0; y <= i; y++) { //left
15             cout << " * ";
16         }
17
18         for (int z = 0; z < i; z++) { //right
19             cout << " * ";
20         }
21
22         cout << endl;
23     }
24
25     return 0;
26 }
```

```
bash
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/te
$ g++ Week\ 1\ Module.cpp -o main.exe && .
* * * * *
* * * * *
* * * * *
* * * *
* * *
*
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/te
$
```

for Number patterns :->

- Just replace the previous stars with either (i, j, k) loop variable according to pattern

1. Square

```
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
```

KEYPOINT:- Replace i with '1' for simple 1 square pattern

OUTPUT:-

```
Week 1 Module.cpp X
Week 1 Module.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n = 5;
6      for (int i = 1; i <= n; i++) {
7          for (int j = 0; j < n; j++) {
8              cout << " " << i << " ";
9          }
10         cout << endl;
11     }
12     return 0;
13 }
```

```
bash X
sharm@DESKTOP-B26G9PU MINGW64 ~/Des
$ g++ Week\ 1\ Module.cpp -o main.e
1 1 1 1 1
2 2 2 2 2
3 3 3 3 3
4 4 4 4 4
5 5 5 5 5

sharm@DESKTOP-B26G9PU MINGW64 ~/Des
$
```

2. Incrementing

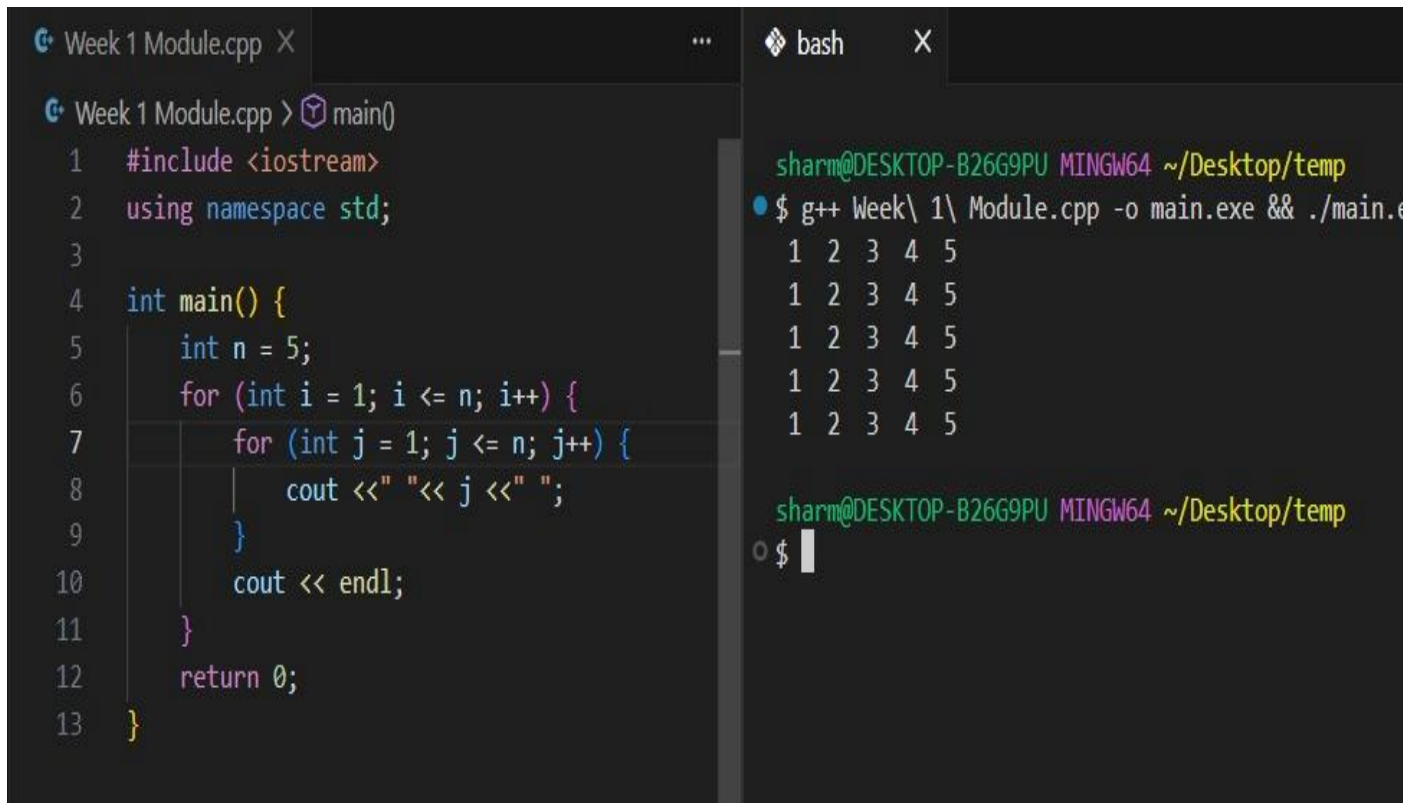
1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5



The screenshot shows a C++ IDE with two panels. The left panel displays the source code for 'Week 1 Module.cpp', and the right panel shows the terminal output.

```
Week 1 Module.cpp X ... bash X
Week 1 Module.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n = 5;
6      for (int i = 1; i <= n; i++) {
7          for (int j = 1; j <= n; j++) {
8              cout << " " << j << " ";
9          }
10         cout << endl;
11     }
12     return 0;
13 }
```

The terminal output on the right shows the execution of the program, displaying a 5x5 grid of numbers 1 to 5, matching the pattern shown in the first part of the image.

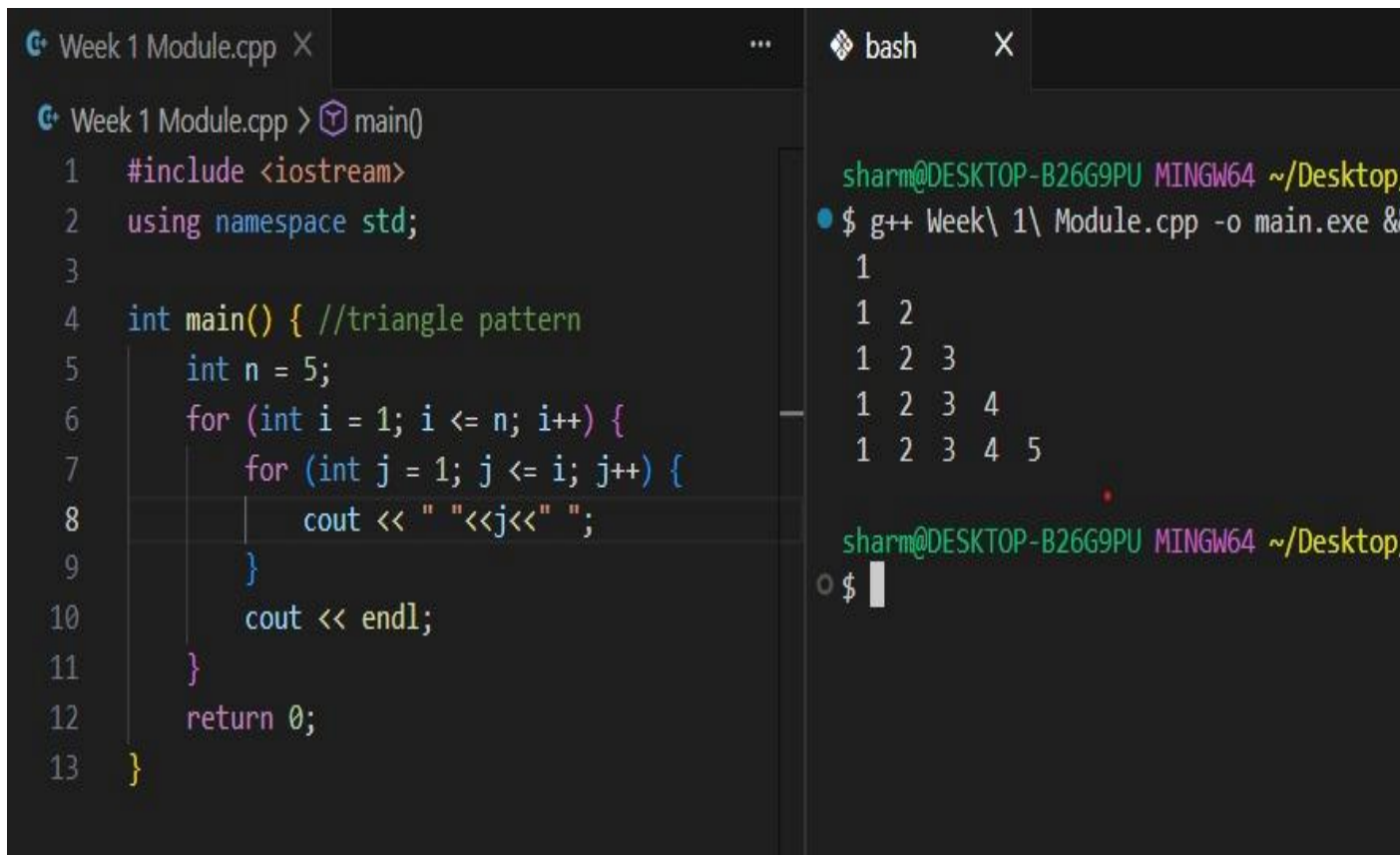
```
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$ g++ Week\ 1\ Module.cpp -o main.exe && ./main.exe
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5

sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$
```

Keypoint:- This problem is similar to Star Problem (square star pattern), just replace '*' with 'j' here and we get our desired numbered pattern

3. Right Triangle

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```



The screenshot shows a C++ IDE with two panels. The left panel displays the source code for 'Week 1 Module.cpp', and the right panel shows the terminal output.

```
Week 1 Module.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() { //triangle pattern
5      int n = 5;
6      for (int i = 1; i <= n; i++) {
7          for (int j = 1; j <= i; j++) {
8              cout << " "<<j<<" ";
9          }
10         cout << endl;
11     }
12     return 0;
13 }
```

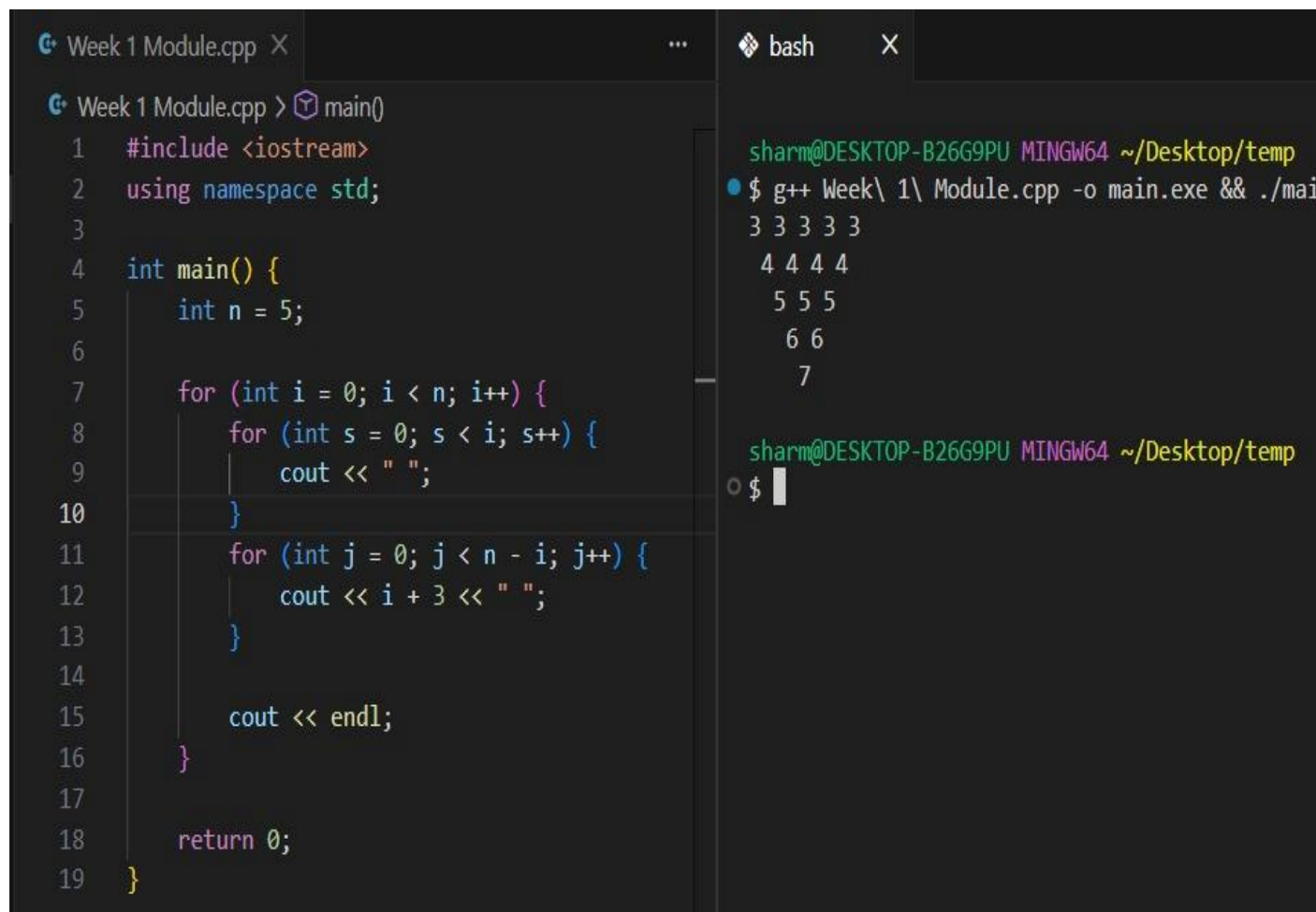
The terminal output on the right shows the execution of the program, displaying the right triangle pattern of numbers:

```
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop
$ g++ Week\ 1\ Module.cpp -o main.exe &
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop
$
```

Keypoint:- This problem is similar to the Triangle star pattern , just replace the “*” with ‘j’ here and we shall get our desired numbered pattern

4. Inverted Diamond

```
3 3 3 3 3
4 4 4 4
5 5 5
6 6
7
```



The screenshot shows a C++ IDE with two panels. The left panel displays the source code for 'Week 1 Module.cpp', and the right panel shows the terminal output of the program.

```
Week 1 Module.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n = 5;
6
7      for (int i = 0; i < n; i++) {
8          for (int s = 0; s < i; s++) {
9              cout << " ";
10         }
11         for (int j = 0; j < n - i; j++) {
12             cout << i + 3 << " ";
13         }
14
15         cout << endl;
16     }
17
18     return 0;
19 }
```

The terminal output on the right shows the execution of the program, displaying the inverted diamond pattern:

```
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$ g++ Week\ 1\ Module.cpp -o main.exe && ./mai
3 3 3 3 3
4 4 4 4
5 5 5
6 6
7

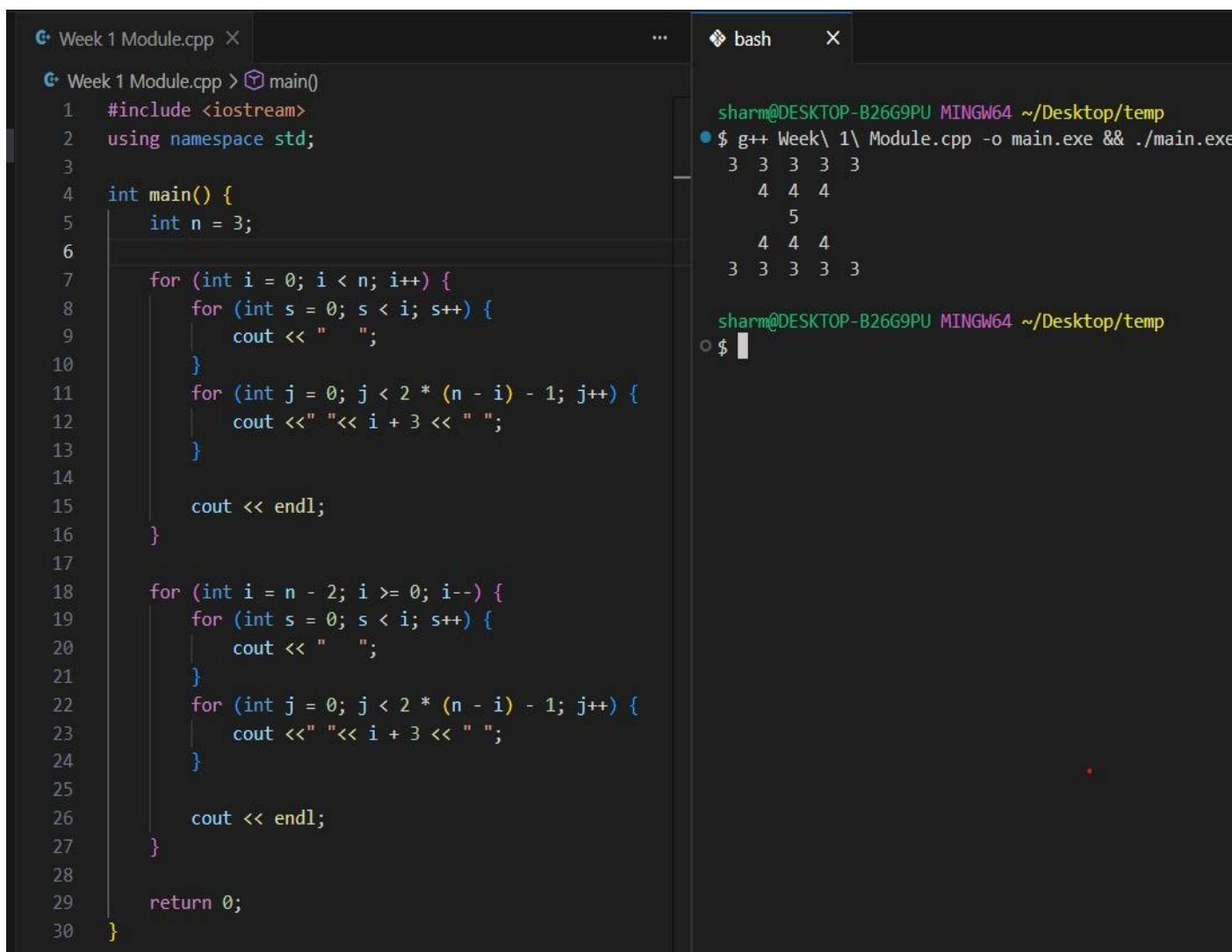
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$
```

Keypoint:- This problem here is also similar to one of the star problems.

5. Sandwich Pattern

```
3 3 3 3 3
 4 4 4
  5
 4 4 4
3 3 3 3 3
```

OUTPUT:-



```
Week 1 Module.cpp X
Week 1 Module.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n = 3;
6
7      for (int i = 0; i < n; i++) {
8          for (int s = 0; s < i; s++) {
9              cout << " ";
10         }
11         for (int j = 0; j < 2 * (n - i) - 1; j++) {
12             cout << " "<< i + 3 << " ";
13         }
14
15         cout << endl;
16     }
17
18     for (int i = n - 2; i >= 0; i--) {
19         for (int s = 0; s < i; s++) {
20             cout << " ";
21         }
22         for (int j = 0; j < 2 * (n - i) - 1; j++) {
23             cout << " "<< i + 3 << " ";
24         }
25
26         cout << endl;
27     }
28
29     return 0;
30 }
```

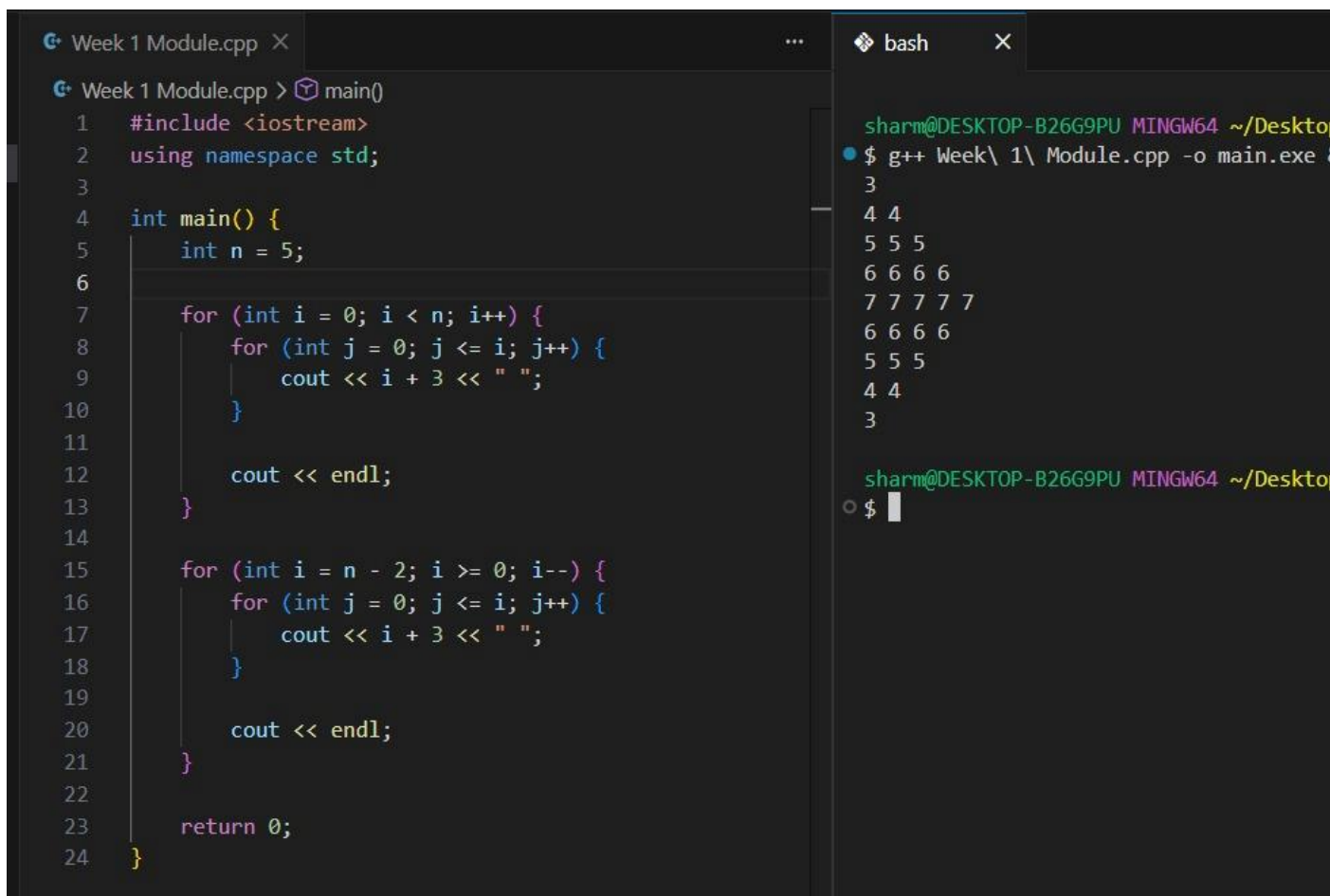
```
bash X
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$ g++ Week\ 1\ Module.cpp -o main.exe && ./main.exe
3 3 3 3 3
 4 4 4
  5
 4 4 4
3 3 3 3 3

sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop/temp
$
```

6. Incrementing Diamond Pattern

```
3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3
```

OUTPUT:-



The screenshot shows a C++ IDE with two panels. The left panel displays the source code for a program that prints an incrementing diamond pattern. The right panel shows the output of the program, which matches the pattern shown in the first block.

```
Week 1 Module.cpp ×
Week 1 Module.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n = 5;
6
7      for (int i = 0; i < n; i++) {
8          for (int j = 0; j <= i; j++) {
9              cout << i + 3 << " ";
10         }
11
12         cout << endl;
13     }
14
15     for (int i = n - 2; i >= 0; i--) {
16         for (int j = 0; j <= i; j++) {
17             cout << i + 3 << " ";
18         }
19
20         cout << endl;
21     }
22
23     return 0;
24 }
```

```
bash ×
sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop
$ g++ Week\ 1\ Module.cpp -o main.exe
3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3

sharm@DESKTOP-B26G9PU MINGW64 ~/Desktop
$
```